

Milestone Supplement: Training, Experiments, Results, and Contributions

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1. Training Methods

Autoencoder. A fully connected autoencoder compresses the 1,525-dimensional jSymbolic/music21 feature vector of each track to a 128-dimensional bottleneck via encoder layers $1526 \rightarrow 512 \rightarrow 256 \rightarrow 128$ (symmetric decoder), ReLU + dropout 0.2. Trained with Adam ($lr=10^{-3}$, batch 256) to minimise MSE for up to 200 epochs; early stopping triggered if $\Delta < 10^{-4}$ over 15 epochs.

KGE (RotatE / ComplEx). Both models are trained for 100 epochs on 807,955 triples (listening edges excluded). Adam optimiser, entity_dim=128, batch 32,768, 256 negatives per positive. Output: 256-dimensional vectors for all 304,110 entities.

HGT. 2 message-passing layers, 4 attention heads, hidden dim 128, dropout 0.1. Track nodes: 384-dim input (128-dim AE || 256-dim RotatE); other types: 256-dim KGE. Per-type MLP head $\rightarrow$ 64-dim ℓ_2 -normalised output. Trained 200 epochs, Adam ($lr=10^{-3}$, wd 10^{-4}) with cosine-annealing. Checkpoint selected by composite Overall Score at $K=10$ on validation.

Loss: logit-adjusted listwise cross-entropy with soft engagement targets $\tilde{p}_{ui} \propto \log(1 + c_{ui})$:

$$s_i = \frac{\hat{\mathbf{u}}^\top \hat{\mathbf{t}}_i}{\tau} + \lambda \log p_i, \quad \mathcal{L} = -\frac{1}{|B|} \sum_{u,i} \tilde{p}_{ui} \log \text{softmax}(\mathbf{s}_u)_i$$

where $\tau=0.1$ and p_i is the Laplace-smoothed marginal listen probability.

KNN-CF. User-user cosine similarity on the ℓ_2 -normalised interaction matrix; $k=6$ neighbours selected by validation sweep.

XGBoost Hybrid. Top-200 candidates per user retrieved via AE-profile cosine similarity, re-ranked with a gradient-boosted LTR model (`rank : ndcg@10`, 300 rounds, early stopping 30) on 300-dim feature vectors.

2. Experiments

- **Autoencoder compression:** transductive 200-epoch run over 7,013 tracks; MSE monitored per epoch.

- **KGE model comparison:** RotatE vs. ComplEx as HGT node initialisers.
- **Init ablation:** full 200-epoch HGT with RotatE-based vs. Gaussian random node features.
- **Logit-adjustment ablation:** 30-epoch sweeps over $\lambda \in \{0.0, 0.05, 0.1, 0.2, 0.5\}$.
- **Multi- K evaluation:** all models at $K \in \{5, 10, 20, 50, 100\}$; Wilcoxon + Friedman tests on a shared 5,000-user subsample.

3. Results and Learning Curves

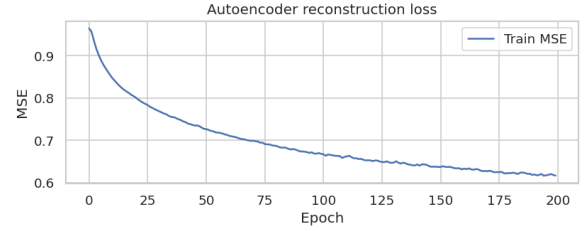


Figure 1. Autoencoder MSE (200 epochs): $0.964 \rightarrow 0.616$. No early stopping triggered; steepest descent in first 75 epochs. RotatE KGE final loss: 0.356; ComplEx: 0.381 (100 epochs each).

Table 1. Test benchmark at $K=10$ (all 284,864 test users). Overall = $0.6 \cdot \text{NDCG} + 0.2 \cdot \text{Cov} + 0.2(1 - \text{PB})$.

Model	Rec.	NDCG	HR	Cov.	PB	Ovrl.
MostPopular	.129	.114	.200	.006	.611	.148
KNN-CF	.103	.106	.169	.916	.130	.421
XGBoost	.038	.044	.072	.667	.118	.336
HGT [†]	.001	.001	.002	.046	.006	.209
HGT (self)*	.111	.071	—	.399	.070	—

[†] Directed-eval mismatch (trained undirected, eval directed).

* Consistent self-eval (undirected inference, same as training).

KNN-CF achieves the best Overall Score (0.421), dominated by its 91.6% catalogue coverage. MostPopular leads on per-user NDCG but concentrates recommendations on 0.6% of the catalogue. Random init marginally outperforms RotatE init for HGT ($\Delta \text{Recall} +0.007$), suggesting KGE features are subsumed by HGT’s own relational learning.

Table 2. HGT ablations at $K=10$. Init: RotatE vs. random (200 epochs). λ sweep (30 epochs).

	Config	Recall	NDCG	Cov	PB
Init	RotatE	.111	.071	.399	.070
	Random	.118	.076	.407	.069
λ	0.00	.125	.081	.006	.321
	0.05	.124	.079	.006	.319
	0.10	.118	.073	.007	.305
	0.20	.110	.068	.015	.280
	0.50	.077	.048	.088	.123

The debiasing term λ trades ranking quality for coverage: $\lambda=0.5$ triples coverage ($0.6\% \rightarrow 8.8\%$) while halving Recall ($0.125 \rightarrow 0.077$). All pairwise Wilcoxon and t -test comparisons are significant ($p < 0.001$).

4. Member Contributions

Pedro Fanica (54346) Pipeline architecture, `nb_env.py` bootstrap, GraphDB upload and SPARQL validation, HGT training loop with logit-adjusted loss, latent-space GMM/UMAP analysis (§13), cold-start persona GUI (§14).

Aarthi Perumpillichira (66404) KG design and construction: OWL ontology schema, `kg_builder.py`, Wiki-data entity alignment, decade temporal chain, rich KG reification, SPARQL query library and GraphDB inspection.

Maja Skwaroń (66083) Data processing pipeline: LakhMSDLinker, user taste profile filtering, `jSymbolic` and `music21` feature extraction, symbolic-feature autoencoder, KNN-CF and XGBoost baselines, evaluation protocol and statistical significance tests.

References